Cross-Domain Slot Filling as Machine Reading Comprehension: A New Perspective

Jian Liu, Mengshi Yu, Yufeng Chen, and Jinan Xu

***Abstract*—With intelligent dialogue systems becoming more and more important in our daily lives, slot filling, one of the most important components of an intelligent dialogue system, has gotten a lot of attention from academia and industry. Despite many advancements in the single-domain learning paradigm for slot filling, leveraging resources from different domains to boost learning for a target domain remains a challenge. In contrast to prior methods that supplemented a sequence labeling model with slot meta-information, we address cross-domain slot filling as a machine reading comprehension (MRC) problem for the first time, where the extraction of slot values is viewed as a question answering process. In the framework above, we present both static and dynamic question generating mechanisms, which have complimentary effects in diverse cross-domain contexts. Furthermore, we devise a dynamic question generation approach that can generate numerous values for a slot at the same time. Finally, we construct a pre-training and fine-tuning training approach that enables us to improve learning by utilizing MRC's resources. We conducted extensive experiments on four datasets to evaluate our approach, and the experimental results clearly justified the advantages of our approach in various cross-domain settings.**

***Index Terms*—Cross-domain slot filling, machine reading comprehension, slot filling.**

I. Introduction

THROUGH natural language interfaces, intelligent dialogue systems (e.g., Amazon Alexa, Apple Siri) can assist users in accomplishing certain tasks, such as reserving a table at a restaurant [1]–[10]. In such a system, *slot filling* is critical for completing the task, which aims to extract the parameter (i.e., slot) values from a user utterance. For example, given a user request "I'd like to book a table at Hat Restaurant for 6 people" in a BOOKRESTAURANT domain, slot filling require determining that "Hat Restaurant" and "6 people" satisfy the values of two pre-defined slot categories, RESTAURANT_NAME and PARTY_SIZE_DESCRIPTION, respectively.

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Previous slot filling research has used a single-domain learning paradigm, in which training and evaluation are limited to a single domain [4], [7], [9], [11]–[14]. Despite many advancements, this paradigm still requires a large amount of labeled data for the domain, which is time-consuming and labor-intensive. Furthermore, approaches based on this paradigm are typically unable to address the practical situation in which new domains emerge following the deployment of a dialogue system [15], [16]. To address the aforementioned limitations, researchers have begun to investigate *cross-domain slot filling*, which aims to connect resources from various domains in order to improve learning [15]–[18]. Previous cross-domain methods usually add slot-related information to a single-domain learning model, such as slot descriptions [15], [17], slot examples [15], and so on [16], but they still require a lot of domain knowledge for model designation. Furthermore, previous methods assume that the source and target domains have similar slots, which makes it difficult to deal with cases where new, previously unknown slots are involved (see Section VI).

In this paper, we propose a new approach to slot filling that is simple, effective, and does not require extensive domain knowledge, but has significant advantages over previous cross-domain adaptation methods. Particularly, we frame the task as a machine reading comprehension (MRC) problem [19]–[21], where the value of a slot is extracted using a question answering process (Fig. 1(b)), as opposed to most previous works viewing slot filling as a sequence labeling problem (Fig. 1(a)). In this formulation, for example, we can construct a prompt question like "What is the name of the restaurant?" to extract the value of RESTAURANT_NAME. This viewpoint naturally leads to a cross-domain training regime: rather than using domain-specific designation to unify the training, we can simply transform labeled data from various domains into this unified question answering and use it to jointly train a model, reducing the need for domain expertise. Second, using the MRC paradigm, we can simply list questions describing the slots and use the MRC model to answer them, effectively addressing the zero-shot cross-domain adaptation [16]. Third, by converting a slot to a question, the semantics of the slot are incorporated into the reasoning process, allowing for a better understanding of the interdependencies between slots and values. Finally, because slot filling and MRC have the same problem formulation, our method can directly use the MRC dataset (e.g., SQuAD [22]) to improve learning. The data sparsity problem in a target domain is greatly alleviated by this cross-task data augmentation.

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The first concern in our approach is how to convert each slot category into a well-formed query question (for example, the slot RESTAURANT_NAME should be recast as "What is the name of the restaurant?"). In our approach, we look at both static and dynamic methods based on templates and back-translation [23], and we show how they complement each other in various cross-domain scenarios (Section III-B). Given a question, our approach generates an answer for the question and, as a result of slot filing, creates a synthesis of all answers. As the foundation, we create a BERT-based MRC model [24] and we devised a dynamic answer generation method that supports generating multiple answers for the same question to fit the case where one slot category corresponds to many values (Section III-C). Finally, we devise a two-step learning paradigm in which the model is first trained on external MRC datasets to gain generic knowledge, and then fine-tuned in the target domain to capture domain-specific knowledge, to effectively leverage out-of-domain resources for training. We show this cross-task data augmentation strategy is especially effective in dealing with data sparsity in a data-low scenario.

We conducted extensive experiments on four benchmarks to validate the effectiveness of our approach: SNIPS [25], ATIS [1], MultiWOZ [26], and SGD [27], taking into account a wide range cross-domain settings. The effectiveness of our approach has been demonstrated by the experimental results. Our approach yields a 7.44 percent absolute gain in F1 over current state-of-the-art methods in a common zero-shot cross-domain setting [28], where a model is trained on all domains except the target one (Section V-A; our approach can yield larger improvements in a few-shot cross-domain setting [15] where the target domain is assumed to have a few examples (Section V-B) Our approach outperforms state-of-the-art methods by 10.4% in F1 on average in a more pure zero-shot cross-dataset evaluation [16], where a model is trained on one dataset but tested on others (Section V-C). In addition, we conducted a thorough investigation into the underlying causes of our approach's effectiveness. This work's code can be found at https://github.com/mengshiY/RCSF.

Our contributions are summarized as follows:

- We take a different approach to cross-domain slot filling in this article, framing the problem as a machine reading comprehension (MRC) problem rather than a sequence labeling problem. To our knowledge, this is the first study of cross-domain slot filling through the lens of MRC, which demonstrates significant advantages over previous cross-domain adaptation methods.
- We propose static and dynamic question generation strategies, which demonstrate complementary effects in various cross-domain scenarios; we devise a dynamic question generation strategy, which supports simultaneously generating multiple values for a slot. Furthermore, we introduce the pre-training and fine-tuning training procedure to enable the use of resources in MRC for data augmentation.
- On four datasets with various cross-domain settings, we provide empirical results validating the effectiveness of our approach. Furthermore, we conducted in-depth analyses to determine the source of effectiveness.

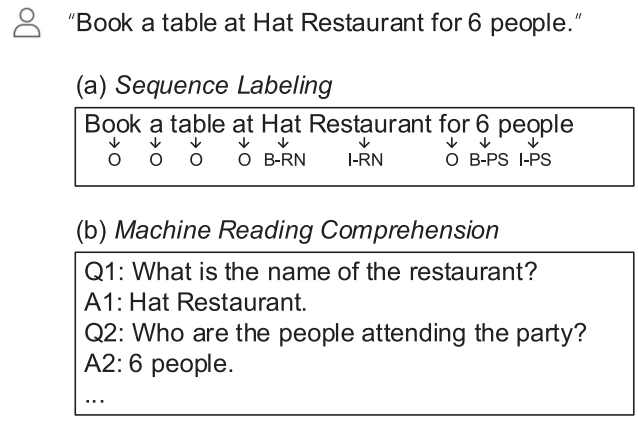

Fig. 1. Comparison of the views of sequence labeling and MRC for slot filling (Here RN and PS stand for two slots RESTAURANT_NAME and PARTY_SIZE_DESCRIPTION respectively). In sequence labeling, a model assigns each word with a slot label in BIO format to identify value spans. By comparison, in our MRC perspective, each slot is cast as a well-designed question, and an MRC model is adopted to predict the answer span.

II. Related Work

A. Slot Filling in Single-Domain Setting

Slot filling is an important component in task-oriented dialogue systems, which strive to recognize the values of slots in order to complete the task. The most popular approach for slot filling is to use a single-domain environment and frame the problem as a sequence labeling problem. Under this learning paradigm, traditional approaches address slot filling usu probabilistic models such as Hidden Markov Models (HMMs) [29] and Condition Random Fields (CRFs) [30], with the goal of capturing the joint probability distribution of the utterance tokens and their slot categories. In the age of deep learning, researchers have begun to use neural networks-based approaches for slot filling. To name a few, [31] design a Recurrent Neural Networks (RNNs) based model for slot filling, regarding the discriminative information conveyed by named entities, future words, syntactic features and word-class information as features for learning [32] compare and identify the impact of different RNNs implementations, such as Elman RNNs [33] and Jordan RNNs [34], for slot filling; [35] propose a slot filling model based on Long Short-Term Memory Networks (LSTMs) [36], demonstrating promising results; [4] and [35] respectively apply recurrent CRFs for slot filling, with a goal to mitigate the labeling bias issue; [37] propose a model to capture the non-local dependency between slot categories to boost learning; [38] design a slot filling model based on bi-directional RNNs and ranking criterion, which can further benefit learning. The latest studies have investigated performing slot filling jointly with intent detection, either in an implicit meaner [6], [39]–[41], or in an explicit manner [9], [42]–[51].

Despite significant breakthroughs, one shortcoming of the single-domain learning paradigm is that it needs obtaining a large number of annotations in a specific domain, which is time-consuming and error-prone. Furthermore, because such a learning paradigm requires all slot categories to be pre-defined, it

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cannot generally address a practical environment in which new domains (with previously unseen slot categories) appear after the deployment of a system.

B. Slot Filling in Cross-Domain Setting

To solve the constraint of single-domain learning, cross-domain slot filling has arisen, with the goal of connecting resources from other domains to enhance learning [15], [17], [28], [52]–[56]. Earlier works have used the shared slot categories as proxies to guide learning [52]–[54], achieving good adaptation performance. While, considering such approaches have limited ability to addressing previously unseen slot categories, recent work has investigated more advanced methods for domain adaptation [15], [17], [28], [55], [56]. For example, several works [15], [28], [56] have considered integrating the definition of slots to enhance learning, noting that similar slot categories usually have similar definitions, and therefore it is possible to recognize the values of unseen slot categories by taking their definitions into consideration. Following the above studies, more advanced approaches have explored integrating specific examples of the slots to enhance learning [17], [55], suggesting that combining with a small number of slot examples can better learn slot semantics. A recent state-of-the-art approach [16] suggests that mining domain-specific linguistic knowledge and pre-trained contextual representations [57] leads to better cross-domain adaptation result.

The progress described above has significantly advanced cross-domain slot filling. However, in order to obtain good performance, such techniques rely on domain-specific designations, which necessitate extensive domain understanding. Furthermore, as demonstrated in Section VI, these approaches struggle with the scenario of entirely unknown slots. Unlike prior methods, we study a novel perspective in MRC for cross-domain slot filling, which is distinguished by its simplicity, efficacy, lack of domain expertise, and strong cross-domain adaptation performance.

C. MRC for Natural Language Processing Tasks

Machine reading comprehension (MRC) is a cutting-edge artificial intelligence (AI) technique that tries to test a machine's capacity to comprehend texts by asking it to answer specific questions about a context [19]–[21]. A recent rush of work has investigated the use of MRC approaches to handle classic natural language processing (MLP) tasks. For example, [58] set up an MRC framework for ten NLP tasks including machine translation, natural language inference, semantic role labeling, and others; [59] frame named entity recognition (NER) as MRC and demonstrate superior performance in handling nested entities; [60] cast relation extraction (RE) as MRC and design a template-based query generation procedure to identify semantic relations between entities; [61] leverage the MRC framework to handle dialogue state tracking task; [62], [63] address event extraction (EE) using MRC techniques. Motivated by the success of the preceding efforts, we provide in this paper a framework based on MRC for cross-domain slot filling, displaying clear improvements over previous approaches based on sequence labeling. Our findings may inspire more research into this potential direction.

III. Methodology

A. Overview

Let $X = \{x_1, x_2, \cdots, x_n\}$ be a user utterance with n words in the slot filling task, and S_Y be a set containing all slot categories corresponding to the domain that X belongs to. The goal of slot filling is to fill each slot category's value with a text span[1] $a = [a_s, a_e]$, where a_s and a_e indicate the start and end positions of a in X. The goal of cross-domain slot filling is to use resources from one or more source domains to boost learning for a target domain.

Fig. 2 depicts an overview of our approach, denoted by SLMRC (Slot Filling via Machine Reading Comprehension). Unlike most previous methods, which framed slot filling as a sequence labeling problem, we take an MRC approach to the task and frame it as a question answering problem. Specifically, given a user utterance, SLMRC converts it into a query question Q_y for each semantic role y. Then, SLMRC configures a BERT-based MRC model [24] to answer the question Qy and predict an answer span a. Finally, as a result of slot filling, SLMRC synthesizes all of the answers. To take advantage of SLMRC for cross-domain adaptation, we devise a pre-training followed by fine-tuning training strategy (as shown in Fig. 2 right), in which we first train the MRC model on external MRC datasets as well as datasets from the source domain to learn out-of-domain knowledge, and then adapt the model on datasets from the target domain to capture domain-specific knowledge. The following sections provide technical information about SLMRC.

B. Query Generation for Slot Category: In our approach, query generation, which acts as a bridge between slot filling and MRC, aims to convert each slot category into a well-formed question. We investigate both static and dynamic strategies, which are depicted below:

a) Query Generation With Slot Definitions: In this strategy, we create a template for question generation called "What is the__?", where the blank is filled with the definition of a slot category. In general, we can deduce a slot's definition from its name. For example, using the above template, we can generate a question such as "What is the restaurant name?" for a particular slot category RESTAURANT_NAME. Where earlier research [15], [28] has supplied more accurate human-written slot definitions for frequently used slot filling datasets (e.g., SNIPS [25] and ATIS [1]), we directly employ these for query creation.

b) Query Generation With Slot Examples: Motivated by previous studies that used concrete examples of a slot to learn slot semantics [15], [17], in our second strategy, we devised a template to integrate slot examples into the question: "What is the like [*e1*] or [*e2*]?", where [*e1*] and [*e2*] are filled with concrete examples of a slot category. When examples of slots

[1] In practice, a slot can have zero, one, or many values. For the sake of brevity, we assume that one slot category corresponds to one and only one span. This is addressed in Section III-C.

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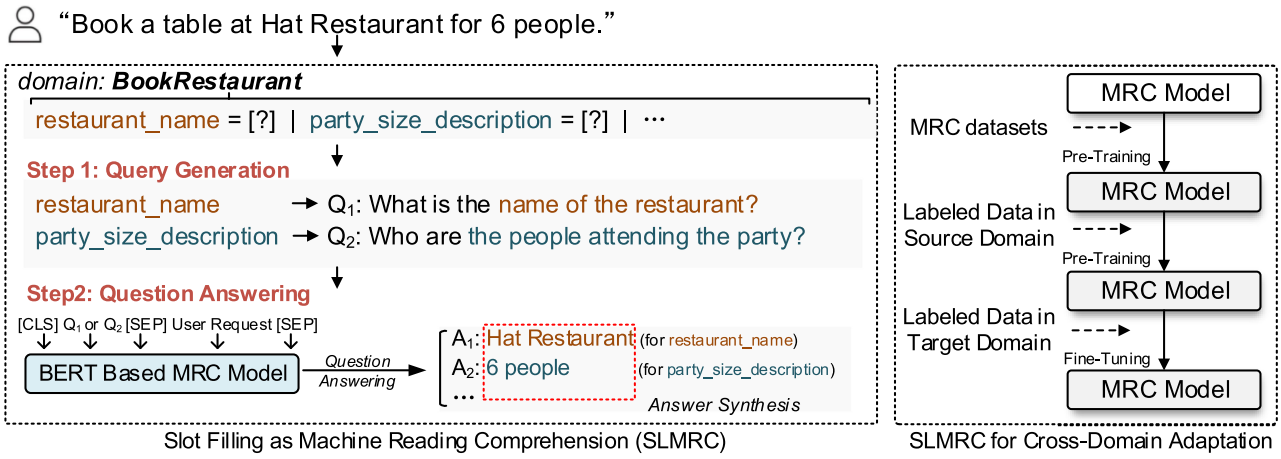

Fig. 2. Left: The overview of our approach SLMRC (Slot Filling via Machine Reading Comprehension). Given a user request in a domain (in this case, BOOKRESTAURANT), SLMRC first converts each associated slot into a query question. Then, as the ultimate result, SLMRC applies a BERT-based MRC model to answer each question and synthesizes all of the answers. Right: The training procedure of using SLMRC for cross-domain adaptation.

are accessible (e.g., SNIPS [25] and ATIS [1]), we use them directly for question generating. In other cases, we use the following procedure to obtain two slot examples of a slot: 1) We generate a huge candidate set that includes all of the noun trunks identified by spacy[2] on English Wikipedia entries. 2) We choose two noun trucks from the candidate set that have the best similarity scores in terms of fastText embeddings [64] with the slot name, as examples. For example, using this strategy, we identify two examples, "a bus stop" and "home", for the DESTINATION slot in the MultiWOZ corpus's TAXI domain.

c) Query Generation With Back-Translation: Despite their simplicity, the above methods are static strategies; they can only generate rigid questions and may introduce noises to learning. For instance, the question "What is the restaurant name?" produced by *Slot Definition* breaks grammatical norms and appears seldom in texts — such noisy inquiries may hamper learning, particularly when using external MRC datasets for domain adaptation, which use natural questions as queries. To address the aforementioned issue, and inspired by prior methods that use back-translation techniques for text de-nosing, [23], we offer a back-translation-based query generation strategy that can modify ill-formed inquiries to make them more natural. In particular, in our approach, the generated question is first translated into a pivot language, and then re-translated back into the original language, considerably improving the question's quality. For example, the method can successfully rephrase the ill-formed inquiry "What is the restaurant name?" as the more natural "What is the name of the restaurant?".

Table I compares the questions generated by the three methodologies discussed above for two specific slot categories.

TABLE I
COMPARISON OF DIFFERENT QUERY GENERATION STRATEGIES

User Request: I'd like to book a table at Hat Restaurant for 6 people.	
Slot Category = RESTAURANT_NAME	
Slot Definition	What is the restaurant name?
Slot Example	What is the restaurant name like the maisonette or the robinson house?
Back-Translation	What is the name of the restaurant?
Slot Category = PARTY_SIZE_DESCRIPTION	
Slot Definition	What is the party size description?
Slot Example	What is the party size description like me or my colleague?
Back-Translation	What is the size of the party?

C. Slot Filling as Answer Generation

Let Q_y represent the generated question for a slot category $y in SY$. We next construct a BERT-based MRC model [24] to predict an answer for Q_y and synthesis all of the answers as the final result. Particular, given Q_y and the user utterance X, we first concatenate them into an extended sequence I:

$$I = [\text{CLS}]\ Q_y\ [\text{SEP}]\ X$$

where [CLS] and [SEP] are special tokens used in BERT. Then we consider I to be the BERT model's input and compute a joint representation for I, which corresponds to the output vector of the BERT's final layer. The component corresponding to X in the joint representation is denoted by $\boldsymbol{H} \in \mathbb{R}^{n \times d}$, where n is the length of X and d is the dimension of BERT. The following step is to predict an answer span for Q_y, based on $\boldsymbol{H}$, which is set as the value of slot y.

In traditional MRC problems, one question usually corresponds to zero or one answer but rarely multiple answers, however in slot filing, a slot might correspond to many values (13 percent of cases in SNIPS [25]. To overcome this issue, we develop an adaptive answer generation system comprised of two processes:

1) Candidate Start and End Position Prediction: In this process, we create two token-wise classifiers — one for predicting the candidate start positions of the answer and the other for predicting the candidate end positions. Particularly, given $\boldsymbol{H}$, the start position classifier computes a probability vector $\boldsymbol{p}_{start} \in$

[2]https://spacy.io/

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$\mathbb{R}^n$, where the ith element $\boldsymbol{p}_{start}^{(i)}$ reflects the probability of i being the answer's start position. The precise calculation is as follows:

$$\boldsymbol{p}_{start} = \sigma(\boldsymbol{H}\boldsymbol{w}_{start} + \boldsymbol{b}_{start}) \tag{1}$$

where σ is an element-wise sigmoid function and $\boldsymbol{w}_{start} \in \mathbb{R}^d$ and $\boldsymbol{b}_{start} \in \mathbb{R}^d$ are model parameters. We use a similar method to predict the candidate end positions, which are represented by $\boldsymbol{p}_{end} \in \mathbb{R}^n$:

$$\boldsymbol{p}_{end} = \sigma(\boldsymbol{H}\boldsymbol{w}_{end} + \boldsymbol{b}_{end}) \tag{2}$$

where $\boldsymbol{w}_{end} \in \mathbb{R}^d$ and $\boldsymbol{b}_{end} \in \mathbb{R}^d$ are additional model parameters to be optimized.

2) Adaptive Answer Generation: We design an adaptive answer generation mechanism that allows us to generate numerous answers at the same time. Based on $\boldsymbol{p}_{start}$ and $\boldsymbol{p}_{end}$, we sort all start and end positions by their predictive probabilities, and then select the top N start and end positions[3] to create two ordered lists, denoted by C_{start} and C_{end}. Then, for each possible span $[s, e]$ with $s \in C_{start}$ and $e \in C_{end}$, we compute a sum probability $p([s, e]) = \boldsymbol{p}_{start}^{(s)} + \boldsymbol{p}_{end}^{(e)}$, in which $\boldsymbol{p}_{start}^{(s)}$ and $\boldsymbol{p}_{end}^{(e)}$ denote the sth and e^{th} elements in $\boldsymbol{p}_{start}$ and $\boldsymbol{p}_{end}$, respectively. Finally, we rank all feasible spans by their sum probabilities and propose a greedy strategy to choose non-overlapping spans with the highest probabilities as the final result of slot filling. In practice, we've found that incorporating a span-related score into the sum probability improves the performance. We specifically train a binary classifier (using golden value spans as positive examples and randomly generated spans as negative examples) to predict whether a span is legal or not, and then revise the sum probability as $p([s, e]) = \boldsymbol{p}_{start}^{(s)} + \boldsymbol{p}_{end}^{(e)} + \mathcal{S}_{[s,e]}$, where $\mathcal{S}_{[s,e]}$ denotes a score awarded by the classifier for $[s, e]$.

D. Cross-Domain Adaptation With SLMRC

We design a pre-training followed by fine-tuning regime for cross-domain adaptation that includes three sub-stages:

1) The Cross-Task Pre-Training Stage: We pre-train the MRC model on external MRC datasets at this step to gather generic information. A training instance is denoted by (X, q, a), which represents a context, query, and answer triple. To train the model, we develop the following loss function:

$$\mathcal{L}(\Theta) = -\sum\nolimits_{(X,q,a)} \mathcal{L}_s(X, q, a) + \mathcal{L}_e(X, q, a) \tag{3}$$

where Θ denotes the overall parameters of the MRC model; $\mathcal{L}_s(X, q, a)$ and $\mathcal{L}_e(X, q, a)$ signify the element-wise entropy losses for predicting the start and end positions of a:

$$\mathcal{L}_s(X, q, a) = \sum_i \boldsymbol{a}_s^{(i)} \log \boldsymbol{p}_s^{(i)} + (1 - \boldsymbol{a}_s^{(i)}) \log(1 - \boldsymbol{p}_s^{(i)}) \tag{4}$$

$$\mathcal{L}_e(X, q, a) = \sum_i \boldsymbol{a}_e^{(i)} \log \boldsymbol{p}_e^{(i)} + (1 - \boldsymbol{a}_e^{(i)}) \log(1 - \boldsymbol{p}_e^{(i)}) \tag{5}$$

where i ranges over each position in X; $\boldsymbol{a}_s$ and $\boldsymbol{a}_e$ are two (one-hot) sequences that represent the golden start and end positions of a in X; $\boldsymbol{p}_s$ and $\boldsymbol{p}_e$ are two vectors representing probability of each position being the answer's start and end positions, which are computed by the MRC model using Equation (1) and Equation (2); $\boldsymbol{y}_s^{(i)}$ and $\boldsymbol{p}_s^{(i)}$ represent the ith element of $\boldsymbol{y}_s$ and $\boldsymbol{p}_s$, respectively.

2) The Cross-Domain Pre-Training Stage: In this stage, we train the MRC model on source-domain datasets to learn out-of-domain knowledge. To be more specific, we first transform the original labeled data into the MRC formulation — (X, Q_y, a), where X denotes a user request, Q_y denotes a question for a slot y, and a denotes the slot value. Then, for optimization, we use a loss function similar to Equation (3):

$$\mathcal{L}(\Theta) = -\sum\nolimits_{(X,Q_y,a)} \mathcal{L}_s(X, Q_y, a) + \mathcal{L}_e(X, Q_y, a) \tag{6}$$

where Θ denotes the MRC model's general parameters When a slot corresponds to more than one value, we create separate training instances for optimization.

3) The In-Domain Fine-Tuning Stage: Finally, during this stage of in-domain fine-tuning, we train the MRC model on datasets from the target domain in order to accomplish domain-specific knowledge. For learning, we use a training objective comparable to Equation (6). Note this stage is not done in the zero-shot cross-domain setting; rather, we leverage the pre-trained MRC to address slot filling in the target domain directly.

IV. Experimental Setups

A. Datasets

We ran extensive experiments on four datasets to validate the effectiveness of our approach:

SNIPS [25]: A publicly available dataset widely adopted for slot filling evaluations [9], [29], [31], [44]–[46]. This dataset defines 39 different slot categories spanning 7 common intents (e.g., BookRestaurant), with each intent having its own slot categories. For example, the intent AddToPlaylist has four slot categories: playlist, music_item, playlist_owner, and artist. For each intent, approximately 2,000 examples are annotated.

ATIS [1]: Another dataset often used for slot filling analyses, which contains audio recordings of users requesting flight information via an automated airline travel inquiry system [29], [31], [46]. This dataset defines 83 slot categories over 18 intents, but many of them have few utterances. We follow the setting of [16] for data pre-processing, in which intentions with fewer than 100 utterances are grouped into a specific intent called "Others".

MultiWOZ [26]: A dataset originally proposed for dialogue state tracking evaluation. To modify this dataset for slot filling, we pick only user utterances and system messages mentioning any slot and shuffle their orders to minimize potential order biases [16]. This yields a dataset with 61 slot categories spread among 11 intents in 7 domains. Following [16], we also classify intents with fewer than 1% of the dataset's utterances (650 samples) as 'Others'.

[3] N is a hyper-parameter tuned on the development set.

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SGD [27]: A recently released dataset proposed in the Dialog System Technology Challenge.[4] This dataset defines 240 slot categories spanning 20 domains and 46 intents. To modify the dataset for slot filling, we follow [16] to filter the dataset and keep only cases with single-turn utterances. We also combine domains with fewer than 1% of the utterances (1850) into a separate domain called "Others".

Given that SNIPS, ATIS, and MultiWOZ do not have a sufficient number of domains, we do the splits for these three datasets based on intents rather than domains, as in earlier works [15]–[17].

B. Evaluation Settings

We explored a wide range of cross-domain situations to test our method, as shown below:

Zero-Shot Cross-Domain Adaptation: This is a commonly utilized scenario in prior publications [15]–[17], [28] to analyze cross-domain slot filling. In this configuration, a model is trained on all domains except the target one, which is used for evaluation. The term "zero-shot" refers to the fact that the model is trained using no training data from the target domain. The above process is repeated for every domain in a dataset.

Few-Shot Cross-Domain Adaptation: This setting is similar to the previous one, but it permits the target domain to have a limited number of training instances for model training [15]–[17], [28]. In this few-shot cross-domain scenario, a model should be able to combine out-of-domain and in-domain knowledge to achieve good performance, as opposed to zero-shot cross-domain adaption.

Zero-Shot Cross-Dataset Adaptation: This is another zero-shot setup given by [16] for cross-domain slot filling, in which a model is trained on one dataset but tested on others. For example, we may train a model on the SNIPS dataset but test it on the ATIS, MultiWOZ, and SGD datasets. Because different datasets usually have distinct schema (i.e., they usually have different domains, intents, and slot categories), this "in the wild" scenario most closely matches a real-world evaluation, which can better test a model's ability for cross-domain adaptability in practical use.

In addition to the above settings, we include a more pure zero-shot setting in our discussion, where we only allow the MRC model to train on labeled data in the MRC domain, but any labeled data in the slot filling task. The empirical outcomes of this situation can illustrate the extent to which external MRC resources can aid in slot filling.

C. Baseline Models

We compare our approach with the following methods:

Concept Tagger (CT) [28]: This approach establishes a sequence labeling framework based on LSTMs [36] and concatenates character, word, and slot representations (i.e., slot type embeddings) as essential features for slot filling. The model's cross-domain capability is mostly attributed to the addition of slot representations.

Robust Zero-Shot Tagger (RZT) [17]: This model, like CT, constructs a sequence labeling framework based on bidirectional LSTMs and uses letter and word representations as basic features for slot filling, but it also uses specific examples of the slot category to boot learning. The findings indicate that combining basic features with slot examples aids in the capture of slot semantics for cross-domain adaptation.

Coarse-to-fine Approach (Coach) [15]: This model handles cross-domain slot filling in a coarse-to-fine paradigm. It first finds candidate spans that tend to be slot values using a coarse sequence labeling framework, and then assigns slot types to these spans by matching their representations with those of each slot category. We compare our strategy to the best model illustrated in [15], namely, Coach+TR, which also uses template regularization to accelerate learning but is denoted simply by Coach for simplicity.

Linguistically-Enriched and Context-Aware Approach (LEONA) [16]: This is a new technique for cross-domain slot filling that augments the sequence labeling framework with domain-independent linguistic characteristics, utterance-slot similarity features, and contextual representation features derived from pre-trained language models (e.g., ELMo [57]). It displays state-of-the-art performance across a wide range of cross-domain tests.

We denote our approach by SLMRC (Slot Filling via Muti-Turn Question Answering) in the following experiments.

D. Implementations

In our approach, we use Google's online translation service[5] to implement the back-translation-based query generating mechanism and use Chinese as the pivot language[6]. We use the BERT-based architecture [24] to construct the MRC model. During the cross-task pre-training stage, we train the MRC model using an external MRC dataset — SQuAD 2.0 [22], attaining 83.1 percent in F1 and matching the state-of-the-art performance [24]. In all other training phases, the learning rate is set to 1e-5 (from [1e-4, 5e-4, 1e-5, 5e-5, 1e-6]); the batch size is set to 64 (from [16, 32, 64, 128]); and the maximum sequence length is set to 128 (from [32, 64, 128, 256]). The aforementioned parameters are fine-tuned using a grid search on the SNIPS development set. The size of the candidate pool in our adaptive question generating method is empirically set at 5 (chosen from 1 to 10). To ensure comparability, we directly employ the code released by prior studies, particularly [15][7] and [16][8]. The code of this work is available at https://github.com/mengshiY/RCSF.

V. Experimental Results

In this section, we go over the experimental results for various cross-domain settings. We employ the basic *Query Generation*

[4]https://sites.google.com/dstc.community/dstc8/home

[5]https://translate.google.com/

[6]We looked into using other languages as the pivot language to implement back-translation, such as Spanish, German, and Japanese, but we saw no improvement. This implies that the pivot language used may not have a significant impact on query generation.

[7]https://github.com/zliucr/coach

[8]https://github.com/abubakar-ucr/LEONA

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TABLE II
F1-SCORES (%) ON SNIPS FOR ZERO-SHOT CROSS-DOMAIN ADAPTATION. HERE SEARCHCW AND SEARCHSE ARE SHORT FOR CREATIVEWORK AND SEARCHSCREENINGEVENT RESPECTIVELY

	Model				
Target Domain ↓	CT	RZS	Coach	LEONA	SLMRC
AddToPlaylist	38.82	42.77	50.90	51.15	**68.70**
BookRestaurant	27.54	30.68	34.01	47.81	**63.49**
GetWeather	46.45	50.28	50.47	**66.77**	65.36
PlayMusic	32.86	33.12	32.01	43.23	**53.51**
RateBook	14.54	16.43	22.06	23.18	**36.51**
SearchCW	39.79	44.45	46.65	46.73	**69.22**
SearchSE	13.83	12.25	25.63	28.72	**33.54**
Average F1	30.55	32.85	37.39	43.94	**55.76**

TABLE III
F1-SCORES (%) ON ATIS FOR ZERO-SHOT CROSS-DOMAIN ADAPTATION, WHERE THE MODELS ARE TRAINED ON ALL DOMAINS (INTENTS) EXCEPT A TARGET ONE FOR EVALUATION.

	Model				
Target Domain ↓	CT	RZS	Coach	LEONA	SLMRC
Abbreviation	41.63	52.52	48.04	64.05	**71.28**
Airfare	65.49	54.10	69.29	**94.92**	94.36
Airline	71.26	63.54	72.12	85.86	**91.11**
Flight	65.30	71.65	80.72	90.70	**92.34**
Ground Service	49.24	64.52	76.41	84.90	**89.21**
Others	48.35	51.69	65.86	83.37	**85.43**
Average F1	56.88	59.67	68.74	83.97	**86.38**

TABLE IV
F1-SCORES (%) ON MULTIWOZ FOR ZERO-SHOT CROSS-DOMAIN ADAPTATION, WHERE THE MODELS ARE TRAINED ON ALL DOMAINS (INTENTS) EXCEPT A TARGET ONE FOR EVALUATION

	Model				
Target Domain ↓	CT	RZS	Coach	LEONA	SLMRC
Book Hotel	45.77	37.39	58.66	64.46	**73.45**
Book Restaurant	32.60	42.00	45.76	62.69	**80.12**
Book Train	47.77	52.69	61.12	70.25	**74.53**
Find Attraction	29.14	34.89	30.29	38.34	**56.89**
Find Hotel	49.33	59.20	72.35	82.22	**84.34**
Find Restaurant	64.20	69.21	76.71	83.38	**88.66**
Find Taxi	14.59	15.87	12.60	18.24	**25.78**
Find Train	63.44	44.06	77.54	88.11	**90.18**
Others	12.05	08.78	12.01	17.21	**25.64**
Average F1	39.88	40.45	49.67	58.32	**67.87**

with Slot Definitions strategy to produce questions for slots for simplicity but without sacrificing generality, and we will analyze the effects of different question generating strategies in Section VI.

A. *Results of Zero-Shot Cross-Domain Adaptation*

We first study the setting of zero-shot cross-domain adaptation [15]–[17], [28], in which a model is trained on all domains except the one being evaluated. The results of several models on SNIPS, ATIS, MultiWOZ, and SGD are shown in Tables II, III, IV, and V, where the span F1 score is reported.

TABLE V
F1-SCORES (%) ON SGD FOR ZERO-SHOT CROSS-DOMAIN LEARNING, WHERE THE MODELS ARE TRAINED ON ALL DOMAINS EXCEPT A TARGET ONE FOR EVALUATION

	Model				
Target Domain ↓	CT	RZS	Coach	LEONA	SLMRC
Buses	49.54	54.43	62.80	69.78	**77.74**
Calendar	50.56	49.08	60.23	74.36	**82.12**
Events	51.81	63.24	54.86	76.19	**76.47**
Flights	48.98	46.62	48.98	59.01	**62.13**
Homes	45.42	71.59	62.35	76.98	**77.81**
Hotels	40.69	56.81	72.16	76.77	**82.33**
Movies	51.00	34.24	55.37	72.85	**75.68**
Music	41.11	60.90	57.86	76.13	**77.11**
RentalCars	41.38	33.99	65.76	73.89	**79.99**
Restaurants	46.20	37.87	71.95	75.74	**76.31**
RideSharing	66.19	53.12	72.73	81.72	**85.03**
Services	63.80	63.81	76.07	81.80	**81.98**
Travel	65.56	64.64	84.03	**92.34**	91.45
Weather	46.05	51.80	60.03	82.23	**84.53**
Others	43.62	53.12	49.21	55.92	**66.67**
Average F1	50.13	53.02	63.63	75.05	**81.03**

The outcomes clearly proved the benefits of our method. Our model SLMRC, in particular, produces the best F1 score on each dataset, outperforming previous state-of-the-art models by 11.82%, 2.41%, 9.55%, and 5.98% in absolute F1. Looking closer, we see that SLMRC earns the top F1 score in practically every domain except a few (such as sc GETWEATHER in SNIPS, which has a prominent pattern), implying that our approach's good cross-domain adaption capacity is generic and universal.

Additionally, we observe that the results on ATIS, MultiWOZ, and SGD are significantly higher than those on SNIPS, where the underlying reason may relate to using the "Others" domain as source domains, which may share slot categories with the target domain. In ATIS, for example, the AIRPORT domain is part of the "Others" domain, which shares slots CITY_NAME and STATE_NAME with the target domain AIRLINE. Given the above issue, our model's significant improvement (+11.82 percent in F1) on SNIPS, which does not have the "Others" domain, firmly illustrates our method's capabilities for cross-domain adaptation.

B. *Results of Few-Shot Cross-Domain Adaptation*

We then investigate few-shot cross-domain adaptation, in which the target domain is considered to have a tiny collection of labeled data [15]. To properly evaluate different models, we run experiments on the less biased SNIPS dataset and use [15]'s setup, which assumes 20 (1%) or 50 (2.5%) labeled samples in the target domain are available. We utilize the code given by [15] for example selection to ensure that the results are comparable. Table VI displays the results of various models for this few-shot configuration.

Clearly, our model SLMRC beats previous state-of-the-art approaches in this setting — it outperforms them by 14.45 percent and 14.53 percent in F1 in cases when 20 (1 percent) and 50 (2.5 percent) training samples are available in the target domain, respectively. The significant performance gap shows that our

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TABLE VI
F1-SCORES (%) ON SNIPS FOR FEW-SHOT CROSS-DOMAIN ADAPTATION. THE NUMBER IN EACH ROW REPRESENTS THE F1 SCORE OF A MODEL ON THE LEFTMOST TARGET DOMAIN. "FEW-SHOT WITH 20 (1%) EXAMPLES" INDICATES THAT ONLY 20 EXAMPLES ARE AVAILABLE IN THE TARGET DOMAIN

	Zero-Shot Adaptation				Few-Shot with 20 (1%) Examples				Few-Shot With 50 (2.5%) Examples			
Domain ↓ Model →	RZT	Coach	LEONA	SLMRC	RZT	Coach	LEONA	SLMRC	RZT	Coach	LEONA	SLMRC
AddToPlaylist	42.77	50.90	51.15	**68.70**	63.18	62.76	63.22	**88.37**	74.89	74.68	76.12	**91.92**
BookRestaurant	30.68	34.01	47.81	**63.49**	50.54	65.97	70.11	**85.56**	54.49	74.82	73.17	**89.64**
GetWeather	50.28	50.47	**66.77**	65.36	58.86	67.89	71.23	**88.83**	58.87	79.64	80.01	**93.90**
PlayMusic	33.12	32.01	43.23	**53.51**	47.20	54.04	59.91	**80.95**	59.20	66.38	68.12	**86.59**
RateBook	16.43	22.06	23.18	**36.51**	63.33	74.68	78.88	**93.35**	76.87	84.62	82.67	**94.06**
SearchCW	44.45	46.65	46.73	**69.22**	63.39	57.19	72.45	**81.30**	67.81	64.56	68.81	**86.23**
SearchSE	12.25	25.63	28.72	**33.54**	49.18	67.38	66.58	**80.46**	74.58	83.85	83.17	**94.22**
Average F1	32.85	37.39	43.94	**55.76**	56.53	64.27	71.10	**85.55**	66.67	75.51	76.42	**90.94**

TABLE VII
F1-SCORES (%) FOR ZERO-SHOT CROSS-DATASET ADAPTION, WHERE A MODEL IS TRAINED ON ONE DATASET BUT TESTED ON OTHERS. THIS CHALLENGING SETTING RESEMBLES MOSTLY A REAL-WORLD CROSS-DOMAIN ADAPTATION SCENARIO. THE IN-DOMAIN PERFORMANCES (F1) OF CT, RZS, COACH, LEONA, AND OUR SLMRC MODEL ON THE FOUR DATASETS ARE: CT (95.2, 94.2, 93.6, 92.8), RZS (95.3, 94.5, 92.9, 91.7), COACH (95.5, 93.7, 92.6, 92.9), LEONA (95.7, 96.3, 93.2, 92.7), AND SLMRC (96.6, 95.3, 94.7, 93.6)

	SNIPS			ATIS			MultiWOZ			SGD		
	ATIS	MultiWOZ	SGD	SNIPS	MultiWOZ	SGD	SNIPS	ATIS	SGD	SNIPS	ATIS	MultiWOZ
CT [28]	8.74	10.99	8.45	5.89	7.25	5.31	6.46	8.78	6.16	14.63	22.90	15.29
RZS [28]	9.15	12.09	10.48	8.19	8.09	9.12	14.96	21.03	8.75	19.05	34.35	21.34
Coach [15]	14.35	11.91	13.01	9.76	9.62	8.71	12.01	17.30	11.02	17.95	33.83	19.03
LEONA [16]	20.80	18.32	16.90	14.36	13.94	13.61	18.47	26.62	16.20	27.61	52.05	28.84
SLMRC	**30.65**	**28.32**	**26.15**	**25.61**	**27.31**	**26.53**	**31.24**	**43.23**	**31.11**	**35.42**	**55.11**	**32.23**

technique can successfully combine out-of-domain knowledge with in-domain knowledge to improve learning. Furthermore, we see that the improvement in this few-shot cross-domain scenario (more than 14.00% in F1 on average) is greater than that in the zero-shot cross-domain setting (11.82% in F1). This also implies that with our method, annotating a small amount of data in the target domain can considerably assist cross-domain knowledge transfer.

C. Results of Zero-Shot Cross-Dataset Adaptation

Finally, we exploit the setting of zero-shot cross-dataset scenario, in which a model is trained on one dataset but tested on another [16]. This setting is more difficult than previous cross-domain settings because the training and testing datasets have a good chance of having completely different domains and slot categories — in this sense, it can be considered a "in the wild" evaluation that closely resembles a real-world cross-domain adaptation scenario. The results of different models for this configuration are shown in Table VII.

According to the results, our strategy consistently surpasses earlier methods by a wide amount. In F1, for example, it enhances the state-of-the-art model LEONA by 3.1 percent on average. Looking closer, we see that our approach achieves a relatively high F1 when using the large and wide coverage SGD dataset for training, and a relatively low F1 when using the small and narrow coverage ATIS dataset for training, implying that the scale and comprehensiveness of the training dataset are important for effective cross-domain adaptation. Finally, as expected, all models perform badly in this zero-shot setting, showing that the current models are not yet suitable for deployment in real-world scenarios, and more explorations in this essential but tough research direction are required.

VI. QUALITATIVE ANALYSIS AND DISCUSSION

We perform further qualitative analysis in this section to better understand the efficacy of our strategy.

A. Results of Addressing Unseen Slot Categories

Addressing previously unforeseen vacancies is a critical part of cross-domain adaptation. Following [15], we propose separating the testing data in the target domain into a Seen and an Unseen group and comparing performance between groups. However, we believe that the separation process proposed in [15] is flawed: if a user utterance contains any slot that has not been seen during training, it is added to the Unseen group; otherwise, it is added to the Seen group — this would result in falsely high results on the Unseen group because it also contains seen slot categories. As a remedy, we provide a new "unseen slot" group (denoted by $\text{Unseen}_{\text{slot}}$), which is a slot level criterion that only takes into account slot categories that were not seen during training. For evaluation, we use zero-shot cross-domain adaptation (Section V-A), and the results of several models on Seen, Unseen, and $\text{Unseen}_{\text{slot}}$ groups are shown in Table VIII.

According to the results, earlier methods perform pretty poorly when addressing unseen slots, particularly on the difficult $\text{Unseen}_{\text{slot}}$ group. The underlying reason is that previous methods rely on domain-specific information to learn slot representations, which requires slots in the source domain to be

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TABLE VIII
THE RESULTS OF ADDRESSING PREVIOUSLY UNSEEN SLOT TYPES. THE UNSEEN GROUP COMPRISES UTTERANCES THAT INCLUDE ANY UNSEEN SLOT; THE SEEN GROUP CONTAINS UTTERANCE IN WHICH ALL SLOTS ARE SEEN. WE ALSO CREATE AN $\text{UNSEEN}_{\text{slot}}$ GROUP, WHICH IS A SLOT LEVEL CRITERION THAT EXCLUSIVELY EXAMINES SLOT CATEGORIES THAT WERE NOT OBSERVED DURING TRAINING

	SNIPS			ATIS			MultiWOZ			SGD		
	Seen	Unseen	$\text{Unseen}_{\text{slot}}$	Seen	Unseen	$\text{Unseen}_{\text{slot}}$	Seen	Unseen	$\text{Unseen}_{\text{slot}}$	Seen	Unseen	$\text{Unseen}_{\text{slot}}$
CT [28]	44.07	27.25	3.47	75.52	48.51	8.90	60.62	30.40	5.78	73.62	39.40	10.01
RZA [28]	47.86	28.01	1.69	81.32	51.43	12.21	66.04	33.01	12.78	75.65	44.78	12.21
Coach [15]	51.73	34.23	11.66	77.42	71.66	24.23	70.34	48.95	13.64	79.96	66.14	11.76
LEONA [16]	63.54	40.06	12.32	95.88	75.24	29.25	77.65	59.62	15.55	91.92	81.67	16.17
SLMRC	**64.41**	**42.76**	**25.44**	**95.49**	**80.12**	**55.78**	**79.12**	**64.57**	**37.98**	**91.29**	**83.66**	**34.98**

TABLE IX
F1-SCORES (%) ON SNIPS FOR ABLATION STUDY. COACH_{B} AND LEONA_{B} AUGMENT THE ORIGINAL COACH AND LEONA WITH BERT REPRESENTATIONS; "*W/O* PT" INDICATES REMOVING THE CROSS-TASK PRE-TRAINING STAGE. SEARCHCW AND SEARCHSE ARE SHORT FOR CREATIVEWORK AND SEARCHSCREENINGEVENT RESPECTIVELY

	Model			
Target Domain ↓	Coach_{B}	LEONA_{B}	SLMRC	SLMRC *w/o* PT
AddToPlaylist	50.51	51.23	68.70	63.21 (-5.49)
BookRestaurant	33.78	46.68	63.49	60.11 (-3.38)
GetWeather	48.24	68.72	65.36	65.23 (-0.13)
PlayMusic	31.17	43.20	53.51	50.16 (-3.35)
RateBook	23.14	25.23	36.51	32.78 (-3.73)
SearchCW	45.89	47.01	69.22	55.17 (-14.05)
SearchSE	26.37	27.99	33.54	30.67 (-2.87)
Average F1	37.39	44.01	55.76	51.77 (-3.99)

TABLE X
ABLATIONS FOR THE FEW-SHOT SETTING WITH 20 EXAMPLES ON SNIPS

	RZT	Coach	LEONA	SLMRC w/o PT	SLMRC
Few	16.89	17.32	17.66	34.20	55.40
CrossD.	32.85	37.39	43.94	51.71	55.76
Few + CrossD.	56.53	64.27	71.10	81.12	85.55

TABLE XI
PERFORMANCE BREAKDOWN

	SNIPS	ATIS	MultiWOZ	SGD
LEONA	43.94	83.97	58.32	75.05
BiDAF w/o PT	50.17	84.43	61.78	77.68
BiDAF w/ PT	55.66	85.17	67.48	80.99
Transformer w/o PT	51.67	84.90	61.88	78.89
Transformer w/ PT	54.65	85.99	67.46	80.95
SLMRC w/o PT	51.71	84.99	62.77	78.91
SLMRC w/ PT	55.76	86.38	67.87	81.03

similar to those in the target domain, which is not the case in $\text{Unseen}_{\text{slot}}$, where the source and target domains do not share any slots. In comparison, our methodology does not have the aforementioned constraint and beats earlier methods by a significant margin. To better understand the effectiveness of our method, consider a unseen slot category PLAYLIST_OWNER in the domain ADDTOPLAYLIST. Coach's F1 score for this slot is a flat 0.0, which always incorrectly assigns a seen slot category PLAYLIST to words in the utterance. In contrast, even though the PLAYLIST_OWNER is not observed during training, our model can create a query "What (Who) is the owner of the playlist?" and extract the answer using an MRC model.

B. Ablation Study

In this section, we conduct an ablation research to determine the source of the improvements. To assess the impact of introducing BERT representations [24], we created two models for comparison: Coach_{B} and LEONA_{B}, which augment Coach and LEONA with BERT representations, for comparison; To investigate the impact of cross-task data augmentation, we created a model that omitted the cross-task pre-training stage, indicated by SLMRC *w/o* PT. Table IX uses SNIPS as an example and compares the performance of several models. From the results, 1) introducing BERT representations is not the main factor for the improvement, as Coach_{B} and LEONA_{B} only show slight advantages over the original models. 2) Cross-task data augmentation is important for improvement: without the pre-training stage, F1 performance reduces by 3.99 percent. 3) While, even without the pre-training stage, our model outperforms previous techniques by a significant margin (7 percent in F1). This implies that the new problem formulation is the primary cause causing the improvement, which clearly demonstrates the efficacy of our method.

Table X shows a more detailed ablation study regarding the few-shot cross-domain evaluation, where we use a set of 20 examples on SNIPS as a case. We adopt two settings, where Few indicates using only the small set of in-domain examples for learning and CrossD indicates using only cross-domain datasets for learning. SLMRC w/o PT is our model without pre-training on the MRC datasets.

According to the results, our approach outperforms other models largely regarding both Few and CrossD, which demonstrates its effectiveness. Moreover, we note our MRC formulation is very data-effective — with only 20 examples, it can achieve 34.2% (55.4% with pre-training) in F1, outperforming previous methods with a large margin. Additionally, pre-training the model on external MRC resources can advance the performance further, in both settings.

Table XI shows a study of performance breakdown exploring where the improvement comes from. We adopt the Zero-Shot Cross-Domain Adaptation setting and use the previous best

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TABLE XII
IMPACT OF ZERO-SHOT ADAPTATION WITH MRC RESOURCES ON SNIPS. SLMRC_{RB} USES ROBERTA [66] OTHER THAN BERT AS THE BACKBONE; SLMRC_{JS} IS AN MRC MODEL TRAINED ON A JOINT SET OF VARIOUS MRC DATASETS

	Experimental Setting		
Target Domain	SLMRC	SLMRC_{RB}	SLMRC_{JS}
AddToPlaylist	18.57	18.05	**20.02**
BookRestaurant	27.24	28.12	**28.80**
GetWeather	23.44	**24.12**	22.02
PlayMusic	27.26	28.66	**29.06**
RateBook	3.21	5.57	**7.12**
SearchCreativeWork	7.38	10.11	**16.53**
SearchScreeningEvent	23.93	25.56	**25.70**
Average F1	18.72	21.67	**24.32**

model LEONA for comparison. For our models, BiDAF uses the same representations of LEONA and adopts the bi-directional attention flow architecture [65] for question answering; Transformer is a model that has the same architecture as BERT but with randomly initialized weights; "w/o PT" indicates excluding the procedure of pre-training on external MRC datasets. According to the results: 1) BiDAF (w/o) PT consistently outperforms LEONA by over 2.0% in F1 — because both of them are trained with the same representations and datasets, we draw a conclusion that the MRC formulation is also effective for non-BERT representations. 2) SLMRC improves Transformer only slightly, which indicates that the BERT representation plays a moderately important role for the improvement. 3) The performance of each model is boosted greatly by pre-training on the MRC datasets, which demonstrates that the two-stage pre-training paradigm is crucial for this MRC formulation.

C. *Impact of Zero-Shot Adaptation With MRC Resources*

The results of the preceding experiments have revealed the importance of cross-task pre-training in our strategy, and to better understand its effectiveness, we undertake a more extensive investigation, attempting to uncover the amount to which MRC resources can be employed to boost slot filling. We assume, in particular, that the potential consequences are a result of both modeling and data, and we create the following models for comparison: 1) SLMRC_{RB}, an MRC model using a more advanced RoBERTa architecture [66] other than BERT as the backbone; it achieves 89.4% in F1 on SQuAD 2.0. 2) SLMRC_{JS}, an MRC model trained on a joint set of various MRC datasets, including NewsQA [67], NarrativeQA [68], TriviaQA [69], and SearchQA [70]. Table XII illustrates the results of applying the above models to the SNIPS test set without any labeled data in the slot filling task for training.

According to the data, our original technique SLMRC achieves an average of 18.72% in F1, indicating the prospective direction of directly applying an MRC model for slot filling. Both SLMRC_{RB} and SLMRC_{JS} outperform our previous model, but the latter is significantly superior; this implies that providing more datasets is more effective than adopting advanced model design. When we look closer, we notice

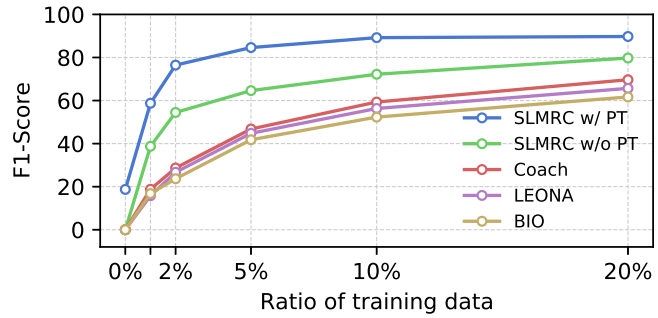

Fig. 3. The impact of cross-task pre-training in a data-low regime, where only a small portion of in-domain data is used to train a model.

that our model produces relatively decent results on domains such as BOOKRESTAURANT and GETWEATHER, but poor results on domains such as RATEBOOK and SEACHCREATIVEWORK. The underlying reason is that the quality of the generated questions could seriously affect knowledge transfer in this zero-shot adaptation setting, and it is generally easy to generate well-formed questions for slots in the former domains (e.g., RESTAURANT_NAME), but not for those in the latter domains (e.g., OBJECT_PART_OF_SERIES_TYPE, OBJECT_SELECT, and BEST_RATING).

Fig. 3 investigates the effect of cross-task pre-training in the data-limited situation, in which we use a tiny fraction of in-domain data to fine-tune the MRC model (in this setting, we do not use datasets in source domain for pre-training). Our model is compared to Coach, LEONA, and a sequence labeling model with BERT representations (denoted by BIO). The performance gap in Fig. 3 clearly confirms the importance of cross-task pre-training in our approach. For example, using only 2% of in-domain training data and cross-task pre-training, our method (SLMRC w/ PT) achieve 80% in F1, whereas earlier methods achieve less than 40% in F1; Even without pre-training, our method (SLMRC w/ PT) is still more data-efficient than earlier methods, indicating the effectiveness of problem formulation.

D. *Comparison of Different Query Generation Strategies*

Because our technique conceptualizes slot filling as a question-answering process, the ways in which queries are constructed may have a substantial impact on the final results. Table XIII studies and compares the performances of different question generation strategies, where Def., Exp., and Back-T indicate strategies using *Slot Definitions*, *Slot Examples*, and *Back-Translation* (shown in Section III-B). We consider two options: Joint Training, in which the model is trained using MRC datasets as well as slot filling datasets (i.e., zero-shot cross-domain adaptation as described in Section V-A), and Zero-Resource Adaptation, which corresponds to the setting described in Section VI-C.

According to the findings, the two advanced question generating (QG) strategies often improve learning, demonstrating the efficacy of our method. In addition, the two tactics have complementary effects: *QG with Back-Translation* outperforms *QG with Slot Examples* in Zero-Resource Adaptation, while

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TABLE XIII
COMPARISON OF DIFFERENT QUERY GENERATION (QG) STRATEGIES, WHERE DEF., EXP., AND BACK-T INDICATE STRATEGIES USING *SLOT DEFINITIONS*, *SLOT EXAMPLES*, AND *BACK-TRANSLATION* RESPECTIVELY

Setting	Target Domain	QG Strategy		
		Def.	Exp.	Back-T
Joint Training	AddToPlaylist	68.70	**70.35**	65.99
	BookRestaurant	63.49	**72.68**	62.05
	GetWeather	65.36	**83.17**	67.80
	PlayMusic	53.51	**53.84**	53.51
	RateBook	36.51	**50.08**	23.67
	SearchCreativeWork	**69.22**	66.59	67.39
	SearchScreeningEvent	33.54	**65.81**	53.20
	Average F1	55.76	**66.08**	56.23
Zero-Resource Adaptation	AddToPlaylist	18.57	18.02	**20.45**
	BookRestaurant	27.24	26.80	**31.07**
	GetWeather	23.44	**28.43**	28.16
	PlayMusic	27.26	28.88	**29.14**
	RateBook	3.21	7.00	**10.05**
	SearchCreativeWork	7.38	**11.53**	11.12
	SearchScreeningEvent	23.93	22.70	**24.76**
	Average F1	18.72	21.15	**23.78**

QG with Slot Examples outperforms *QG with Back-Translation* in Joint Training. A probable explanation is that, in the case of Zero-Resource Adaptation, the model is trained entirely on MRC datasets using natural questions as prompts. Compared with static question generation methods, our dynamic question generation method based on back-translation is effective for text de-noising and generating natural questions, thus resulting in a better knowledge transfer procedure. While in Joint Training, because the labeled data in the slot filling domains is used to train the model, it does not matter whether a method can create natural questions or not, as long as the generated questions are consistent. *QG with Examples*, in this setting, can incorporate the additional information supplied by the slot examples to improve learning. While the dynamic question generation method based on back-translation, which focuses more on generating natural questions, may revise the indicative information conveyed by the slot's name and therefore leads to a degraded performance.

Finally, we note that the fundamental technique, *QG with Slot Definitions*, produces competitive or better results in domains such as PLAYMUSIC and SEARCHCREATIVEWORK, which may be due to the fact that slots in such domains (e.g., PLAYLIST_OWNER) have clear and defined definitions, which are sufficient to create good questions.

E. Error Analysis

To explore the potential limitations of our approach, we choose 200 error cases from the developing set of SNIPS and do an error analysis. From the results, one common error corresponds to the generic roles in the dataset, such as OBJECT_TYPE, OBJECT_NAME and OBJECT_SELECT in the domain SEACHCREATIVEWORK, accounting for 31.9% of the total. It is difficult to produce good questions for such generic roles, and one approach could be to introduce more information to define such slots. Another type of error is created by the fact that our solution uses independent question answering processes and fails to describe their dependencies. This type of error accounts for 18.9% of all cases. Consider the following example. In the domain ADDTOPLAYLIST, there are two slots: PLAYLIST and PLAYLIST_OWNER. Given a user utterance: "Add the artist Jurij Szewczuk to tisha's soul revived playlist", our current method generates two independent questions "What is the playlist?" and "Who is the owner of the playlist?" to extract slot values. In this case, after discovering that "soul revived" is the slot value of PLAYLIST, it is preferable to use a more informative query "Who is the owner of soul revived?" rather than "Who is the owner of the playlist?" to extract the answer. In practice, after incorporating the above slot dependencies into our strategy, we observe a significant performance reduction (3% in F1 for SNIPS for zero-shot cross-domain adaptation) due to error propagation.

VII. CONCLUSION

In this paper, we look at cross-domain slot filling from a different angle: instead of approaching the task as a sequence labeling problem, we frame it as a machine reading comprehension (MRC) challenge. We present both static and dynamic question creation techniques in the framework above, which have complimentary effects in different cross-domain scenarios; we create a dynamic question generation strategy that permits creating multiple values for a slot at the same time. Furthermore, we design a pre-training followed by fine-tuning training approach that allows us to use MRC resources for data augmentation. Extensive testing on four datasets with varied cross-domain scenarios confirmed the efficacy of our technique. In the future, we would investigate better question generating strategies and how to address intent detection and slot filling duties collaboratively in cross-domain scenarios.

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